PROBABLITY BASIC

1.: A die is rolled. What is the probability of getting: (a) An even number (b) A number greater than 4

- A fair die has 6 possible outcomes: $\{1, 2, 3, 4, 5, 6\}$.

Formula:

$$P(E) = \frac{Favourableoutcomes}{Totalpossibleoutcomes}$$

(a) Probability of getting an even number

Even numbers are $\{2, 4, 6\}$.

- Favourable outcomes = 3
- Total outcomes = 6

\[P(\text{Even})=\frac{3}{6}=\boxed{\frac{1}{2}} \]

Answer: $\frac{1}{2}$ or 50%

(b) Probability of getting a number greater than 4

Numbers greater than 4 are $\{5, 6\}$.

- Favourable outcomes = 2
- Total outcomes = 6

\[P(\text{Number}>4)=\frac{2}{6}=\boxed{\frac{1}{3}} \]

Answer: $\frac{1}{3}$ or 33.33%

2.In a class of 50 students: 20 like Mathematics (M) 15 like Science (S) 5 like both subjects What is the probability that a student chosen at random likes Mathematics or Science?

- Given:
- Total students = 50
- Students who like Mathematics, $n(M) = 20$
- Students who like Science, $n(S) = 15$
- Students who like both, $n(M \cap S) = 5$

Formula:

PROBABLITY BASIC

$$P(M \cup S) = \frac{n(M)+n(S)-n(M \cap S)}{Totalstudents}$$

Step 1: Substitute the values.

$$P(M \cup S) = \frac{20+15-5}{50}$$

Step 2: Calculate.

\[=\frac{30}{50}=\boxed{\frac{3}{5}} \]

The probability that a randomly chosen student likes Mathematics or Science (or both) is $\frac{3}{5}$ or 60%.

3.A bag has 3 red and 2 blue balls. If one ball is drawn randomly and is red, what is the probability that the next ball is also red (without replacement)?

- Given:
- Red balls = 3
- Blue balls = 2
- Total balls = 5

Since the first ball drawn is red and there is no replacement, one red ball is removed from the bag.

Step 1: Find the remaining balls.

- Red balls remaining = $3 - 1 = 2$
- Blue balls remaining = 2
- Total balls remaining = 4

Step 2: Calculate the probability of drawing another red ball.

$$P(Nextred) = \frac{Remainingredballs}{Totalremainingballs}$$

\[=\frac{2}{4}=\boxed{\frac{1}{2}} \]

The probability that the next ball is also red is $\frac{1}{2}$ or 50%.

PROBABLITY BASIC

4.The population of a school is divided into 60% boys and 40% girls. If you want equal representation of both genders in the sample, which method should you use: Simple Random Sampling or Stratified Sampling? Why

- Answer: Stratified Sampling

Why?

Stratified Sampling is the best method because we want equal representation of boys and girls in the sample.

- Simple Random Sampling: Students are selected randomly from the entire school. Since 60% are boys and 40% are girls, the sample may not contain equal numbers of both genders.
- Stratified Sampling: Students are divided into two groups (strata): boys and girls. Then students are selected from each group to ensure equal representation.

Example: If we need a sample of 100 students, we can select:

- 50 boys
- 50 girls

This gives equal representation of both genders.

5.: The average height of 1000 students = 160 cm. A sample of 100 students shows an average height = 158 cm. Find the sampling error.

- Given:
- Population mean = 160 cm
- Sample mean = 158 cm

Formula:

$$SamplingError = SampleMean - PopulationMean$$

Calculation:

$= 158 - 160 = -2cm$The sampling error is −2 cm (or 2 cm in magnitude).

This means the sample's average height is 2 cm lower than the population's average height.

-

6. The population mean salary is ₹50,000 with σ = ₹5,000. If we take a sample of 100 employees, what is the standard error of the mean (SEM)?

- Given:
- Population standard deviation (σ) = ₹5,000
- Sample size (n) = 100

Step 1: Write the formula

The standard error of the mean (SEM) is:

$$SEM = \frac{\sigma}{\sqrt{n}}$$

Step 2: Substitute the values

$$SEM = \frac{5000}{\sqrt{100}}$$

$$= \frac{5000}{10} = ₹500$$

Final Answer

\[\boxed{SEM=₹500} \]

Interpretation: The standard error of ₹500 indicates the typical variability of sample mean salaries across repeated samples of 100 employees, assuming the population standard deviation is ₹5,000.

7.: In a group of 100 students:

40 like Cricket ©

30 like Football (F)

10 like both Cricket and Football

Find the probability that a student likes at least one sport

- Given:
- Total students = 100
- Students who like Cricket, $n(C) = 40$
- Students who like Football, $n(F) = 30$
- Students who like both, $n(C \cap F) = 10$

Step 1: Write the formula

The probability that a student likes at least one sport is:

$$P(C \cup F) = \frac{n(C)+n(F)-n(C \cap F)}{Totalstudents}$$

Step 2: Substitute the values

$$P(C \cup F) = \frac{40+30-10}{100}$$

\[=\frac{60}{100}=\boxed{0.6} \]

Final Answer

Probability = 0.6 or 60%.

Therefore, the probability that a randomly selected student likes Cricket or Football or both is 60%.

8.From a deck of 52 cards, two cards are drawn without replacement. What is the probability that both are Aces?

- Given:
- Total cards = 52
- Number of Aces = 4
- Two cards are drawn without replacement.

Step 1: Probability of drawing an Ace first

$$P(FirstAce) = \frac{4}{52}$$

Step 2: Probability of drawing an Ace second

After drawing one Ace, 51 cards remain, including 3 Aces.

$$P(SecondAce) = \frac{3}{51}$$

Step 3: Calculate the probability that both cards are Aces

Using the multiplication rule:

$$P(BothAces) = \frac{4}{52} \times \frac{3}{51}$$

\[=\frac{12}{2652}=\boxed{\frac{1}{221}} \]

Final Answer

\[\boxed{\frac{1}{221}\approx 0.4525\%} \]

Therefore, the probability that both cards are Aces is $\frac{1}{221}$, or approximately 0.4525%.

9.A factory produces bulbs with 2% defective rate. If 5 bulbs are chosen at random, what is the probability that all are non-defective?

- Given:
- Defective rate = 2% = 0.02
- Probability of a defective bulb = 0.02
- Probability of a non-defective bulb = $1 - 0.02 = 0.98$
- Number of bulbs chosen = 5

Step 1: Write the formula

Assuming the bulbs' defect status is independent,

$$P(Allnon - defective) = (0.98)^5$$

Step 2: Calculate

$$P = (0.98)^5$$

$$P = 0.9039207968$$

Final Answer

\[\boxed{P\approx 0.9039=90.39\%} \]

Therefore, the probability that all 5 bulbs are non-defective is approximately 90.39%.

10.: Differentiate between discrete and continuous random variables with examples.

- Difference Between Discrete and Continuous Random Variables

A random variable is a variable whose value depends on the outcome of a random experiment.

Basis	Discrete Random Variable	Continuous Random Variable
Meaning	Takes distinct, countable values.	Takes any value within a given range.
Values	Usually whole numbers, such as 0, 1, 2, 3.	Can include decimals and fractions.
Counting	Values can be counted.	Values are measured.
Examples	Number of students in a class, number of defective bulbs, number of heads in coin tosses.	Height, weight, temperature, time taken to complete a task.
Probability	Probability can be assigned to individual values.	Probability is calculated over intervals; the probability of any exact value is zero.

1. Discrete Random Variable

A discrete random variable takes a countable number of possible values.

Example: Toss a coin 3 times and count the number of heads.

Possible values are:

$$X = \{0, 1, 2, 3\}$$

The number of heads can only be 0, 1, 2, or 3.

2. Continuous Random Variable

A continuous random variable can take any value within a range, including decimal values.

Example: Measure the height of students in a class.

A student's height might be 150 cm, 150.5 cm, or 150.75 cm. Many possible values exist within a range.

Conclusion

- Discrete = Counted (e.g., number of students).
- Continuous = Measured (e.g., students' heights).

Easy trick to remember: If you count something, it is generally discrete. If you measure something, it is generally continuous.